

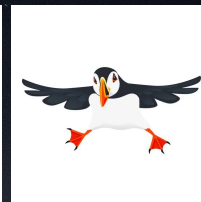
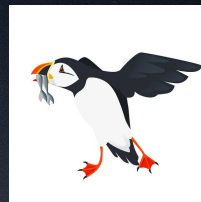
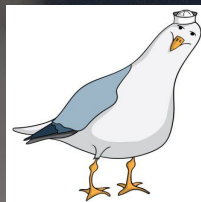
Assessing the impact of temporal resolution on animal behavioural inference

Rebecca Akeresola

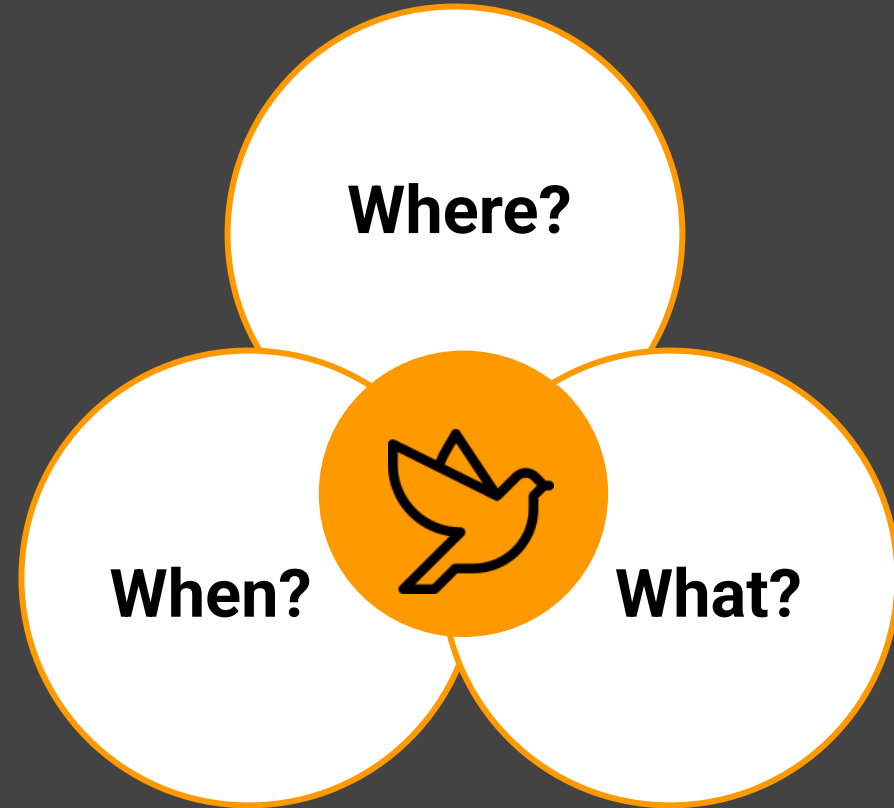
30th April, 2024

MATHBIO SEMINAR

✉ r.a.akeresola@sms.ed.ac.uk



– Background



—

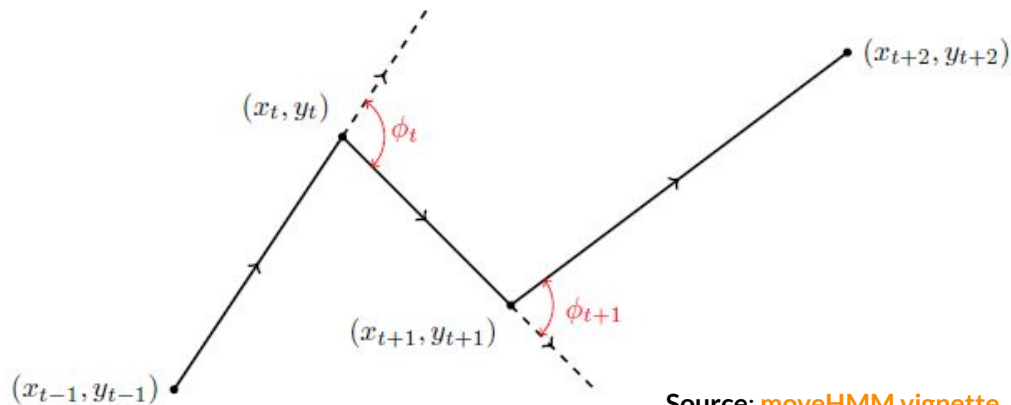
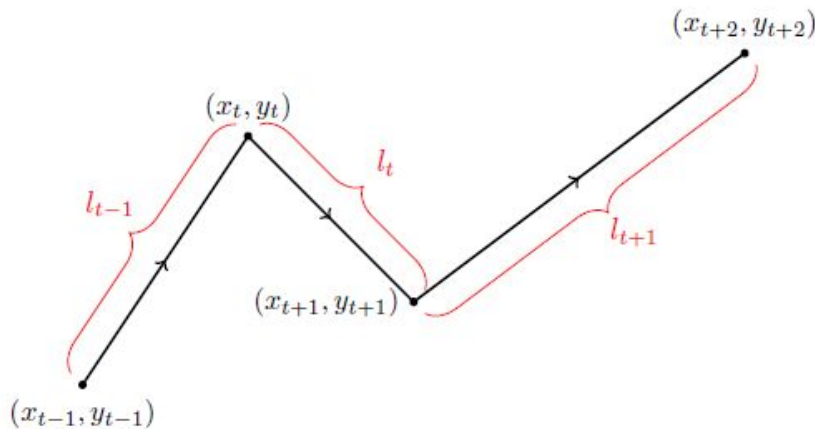
When collecting data on **where** an animal is, we want to answer the question:

“what is the animal doing”



— What?

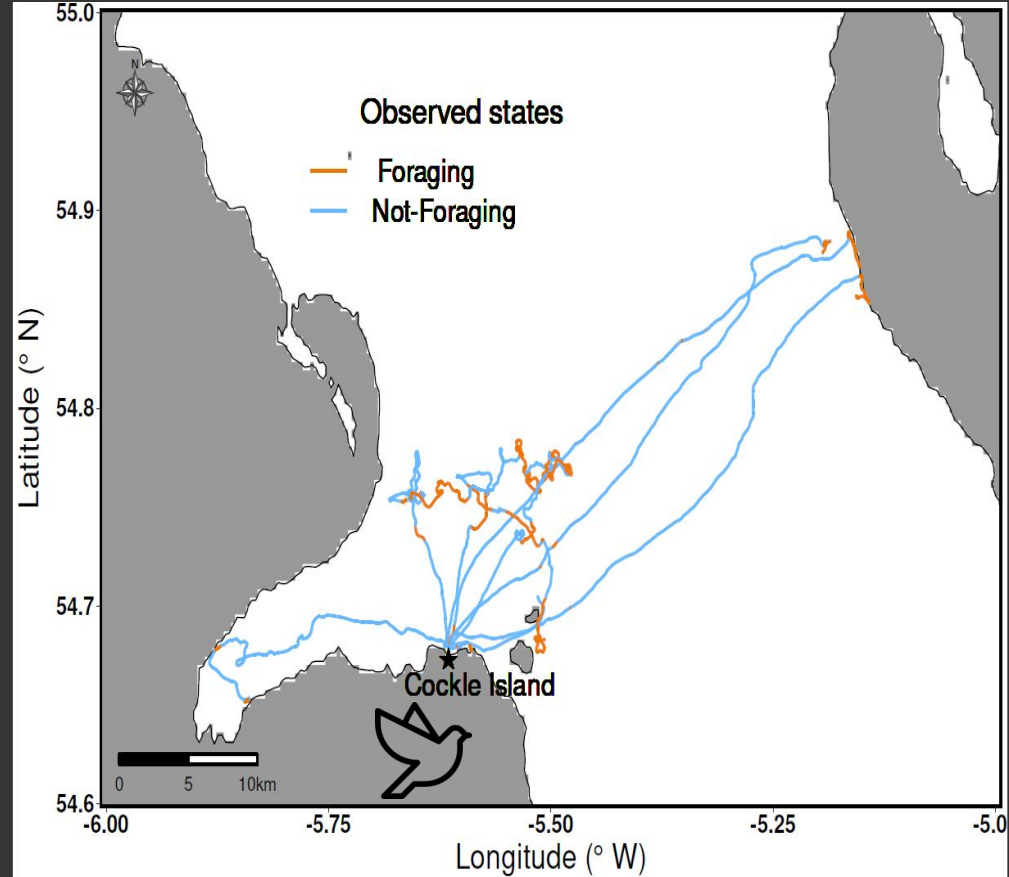
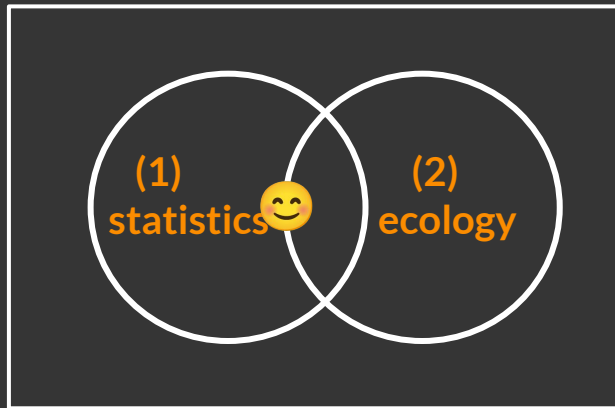
We can convert locations (x,y) into quantities (step length, l , and turning angles, ϕ) that tell us something about animal behaviour



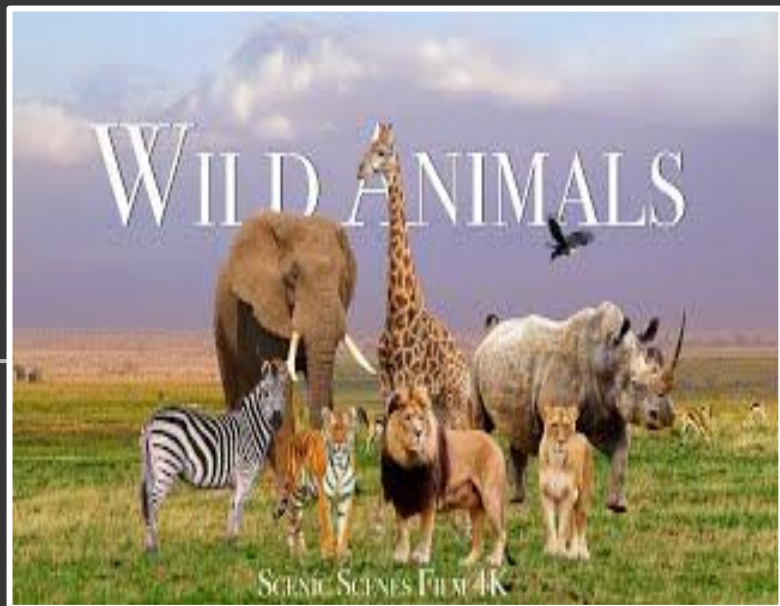
— “What is the animal doing?”

Describe different movement modes or “states” along a track

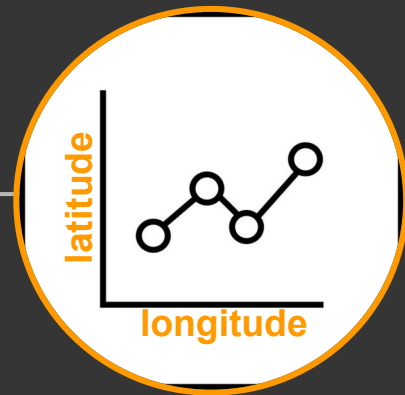
- 1 link state to the conditions at that location
- 2 loosely connect states to behaviours



— When?

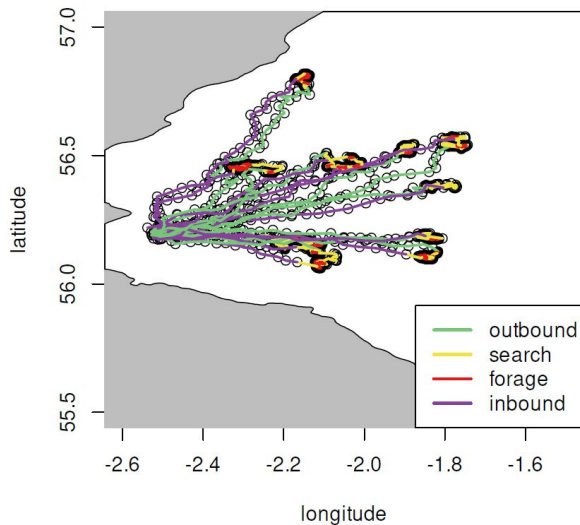


Continuous
time

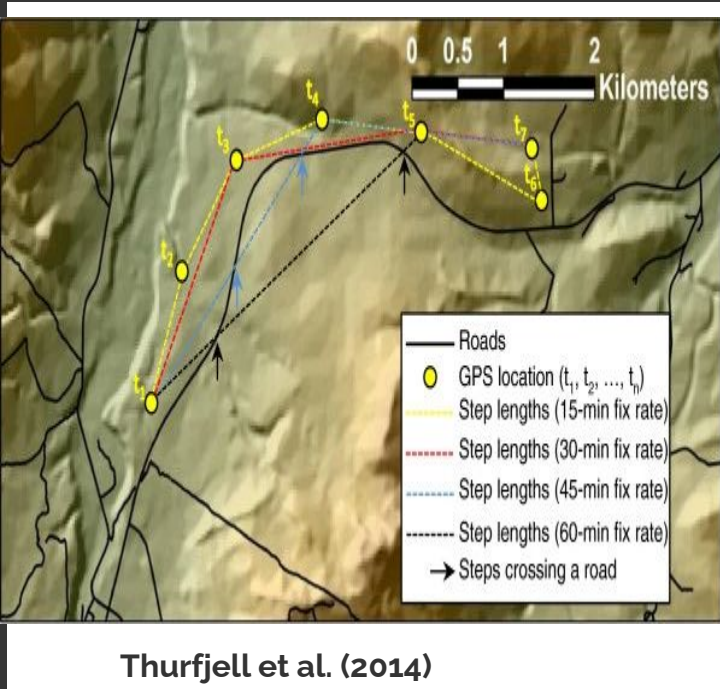


— When?

Movement observed at **discrete times** can either be regular or irregular, and at a fine or coarse temporal resolution



— Problem

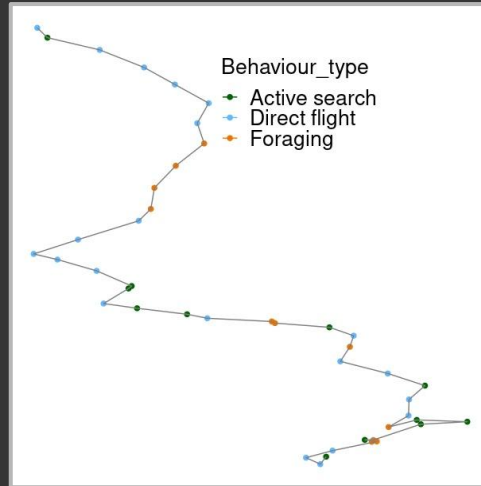
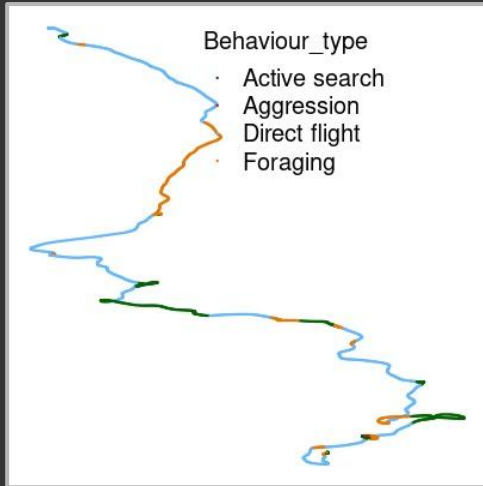


Sampling frequency determines the temporal resolution at which behavioural inferences can be made.

High **sampling frequency** comes at a cost of sampling duration and GPS battery life

Coarse **sampling frequency** may not capture behaviour obtainable at finer temporal scale

— Problem



How do we observe movements
at discrete-time with little or no
loss in underlying behaviour
influencing movement

Fine temporal scale

GPS battery life
Sampling duration

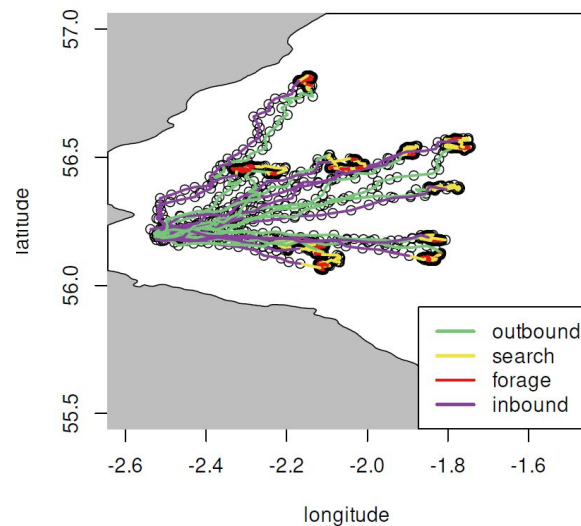
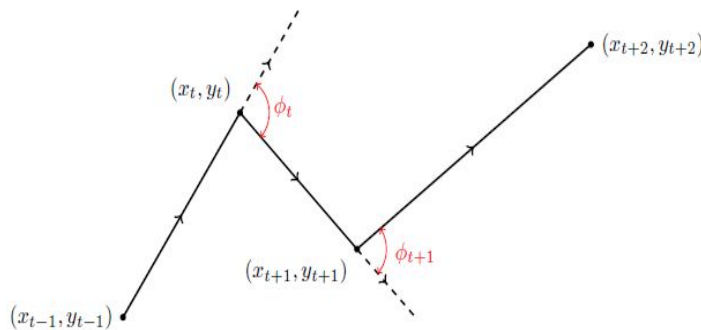
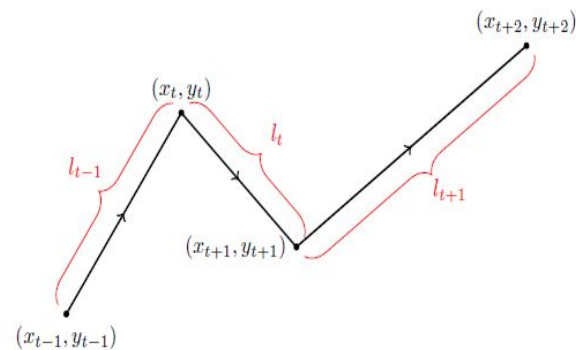
Coarse temporal scale

Miss out on event
obtainable at a fine-scale



Goal

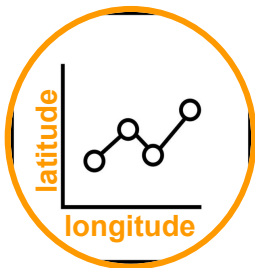
Assess effect of sampling interval, dt , on animal behavioural inference



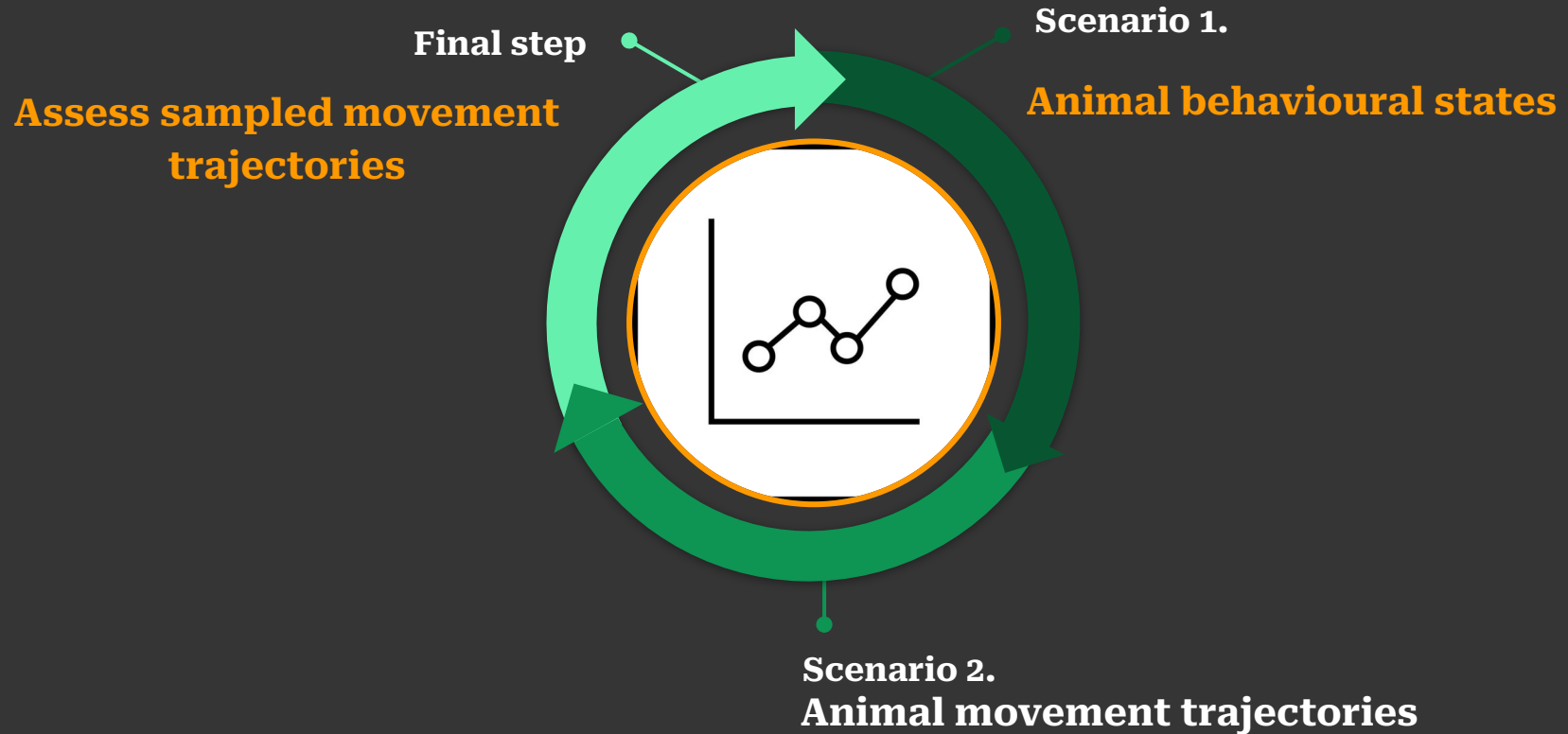
— Nyquist-Shannon sampling theorem (NSST)

NSST describes how to sample a continuous-time signal (or subsample a discrete-time signal) in a way that there is no information loss

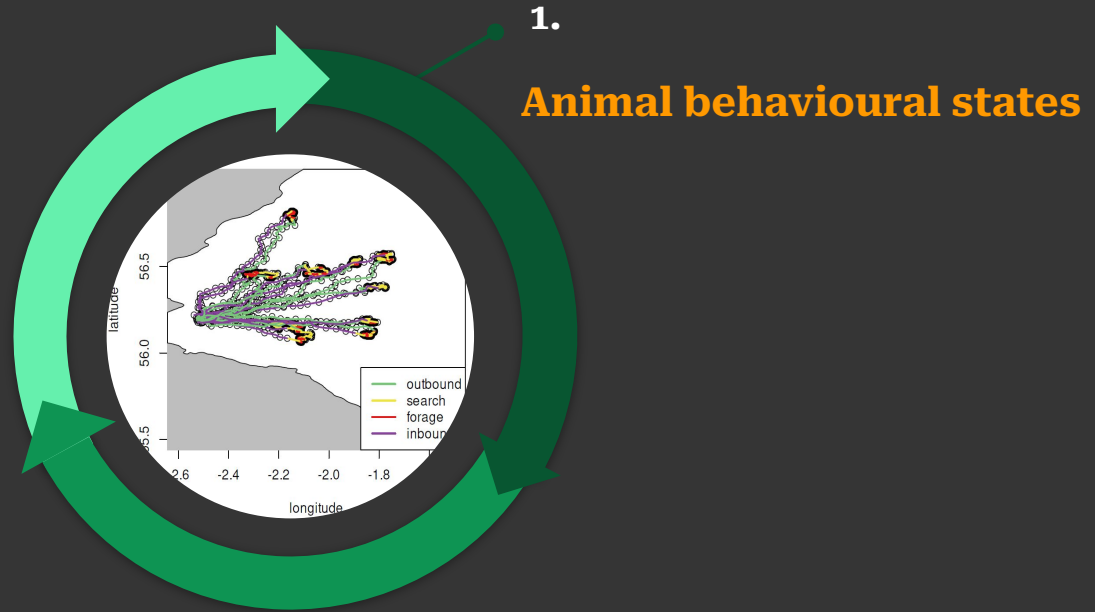
It can be useful for selecting temporal resolutions in the **animal movement trajectories** such that there is no loss of information about the underlying latent behavioural process



— NSST application



— NSST application



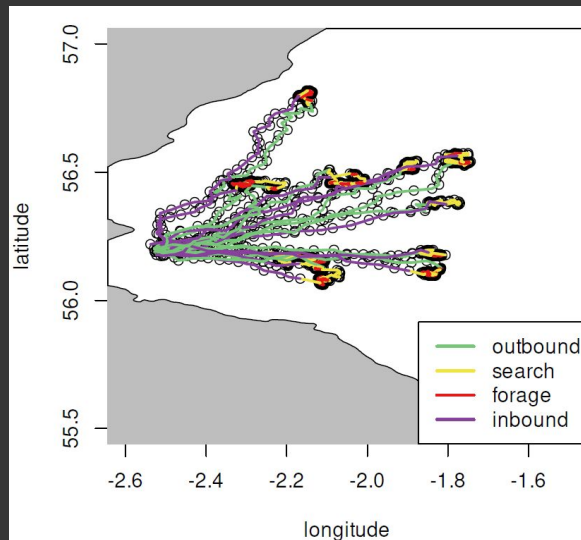
– Nyquist-Shannon sampling theorem (NSST)

Sampling at time interval δt is sufficient to correctly characterize behaviours that last $2\delta t$ or longer

$$\delta t = \min (t_1, t_2, \dots, t_n)$$

n is the number of behavioural states

t is half the minimum time spent in each behavioural bout

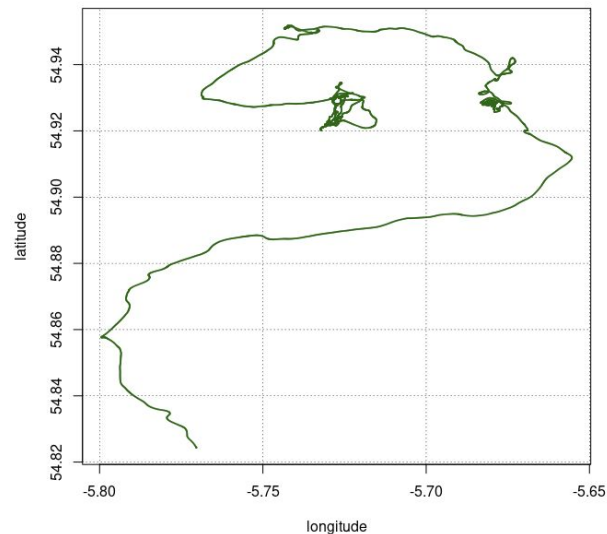


Example: Visual tracking data sampled at $\delta t = 1$ secs

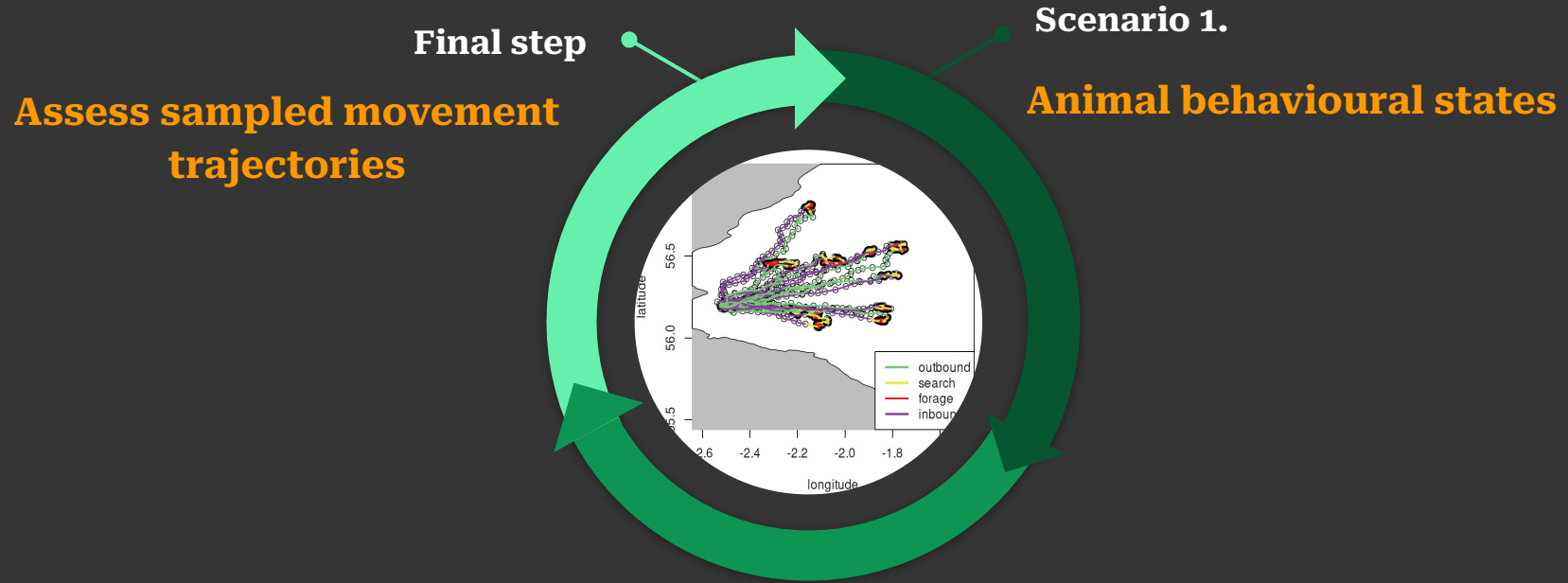
From NSST perspective, subsampling the visual tracking data to correctly characterize behavioural states requires that

$$\delta t = \min(t_1 = 30, t_2 = 20) = 20 \text{ secs}$$

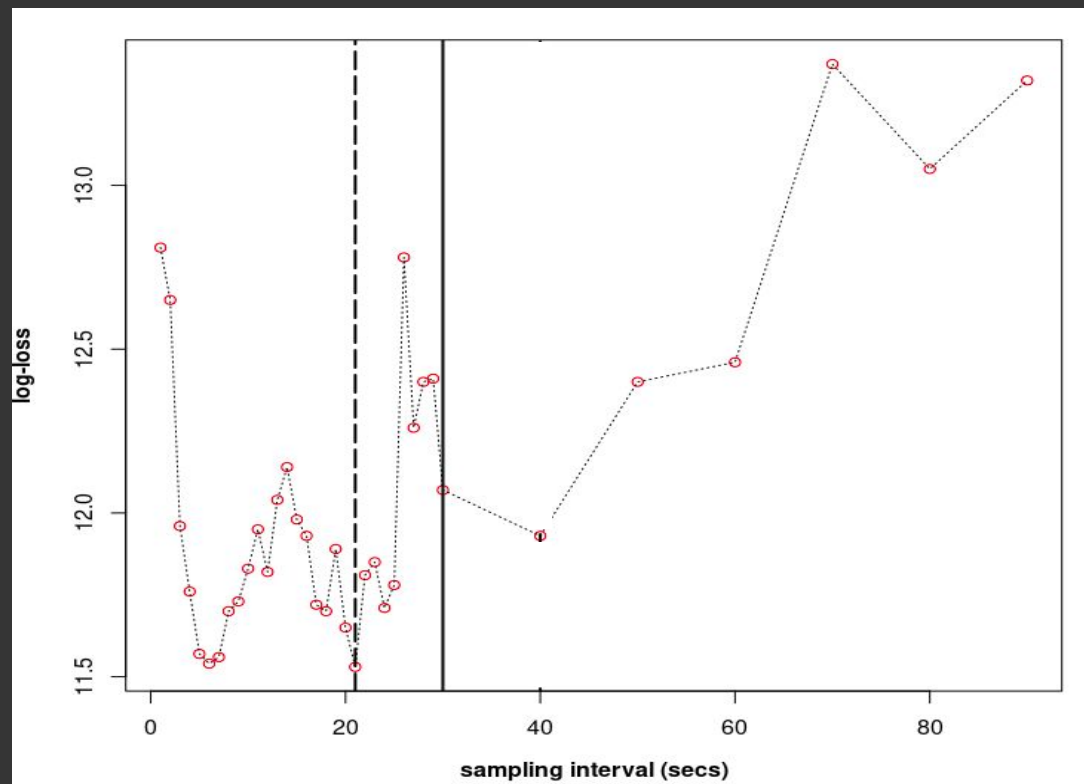
	minimum	maximum
Foraging bout (n=14)		
time (secs)	60	356
$2(\delta t)$	2(30)	2(178)
Not-foraging bout (n=15)		
time (secs)	40	502
$2(\delta t)$	2(20)	2(251)



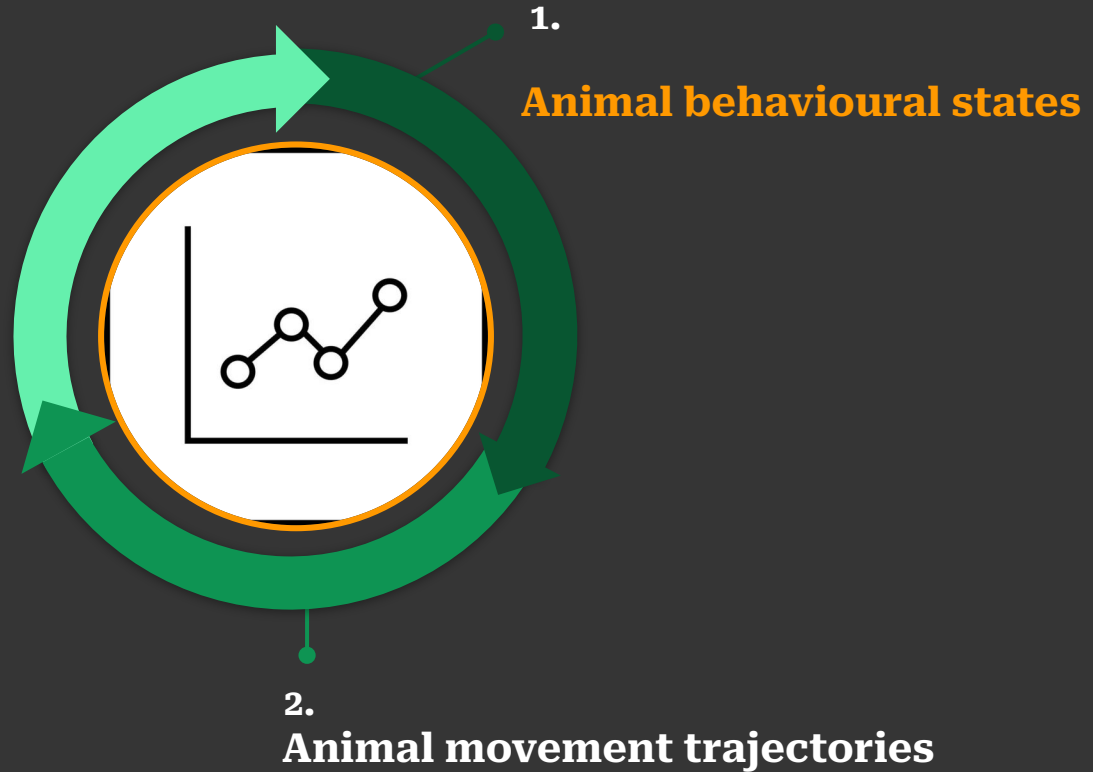
— NSST application



Example: Visual tracking data originally sampled at $\delta t = 1$ secs



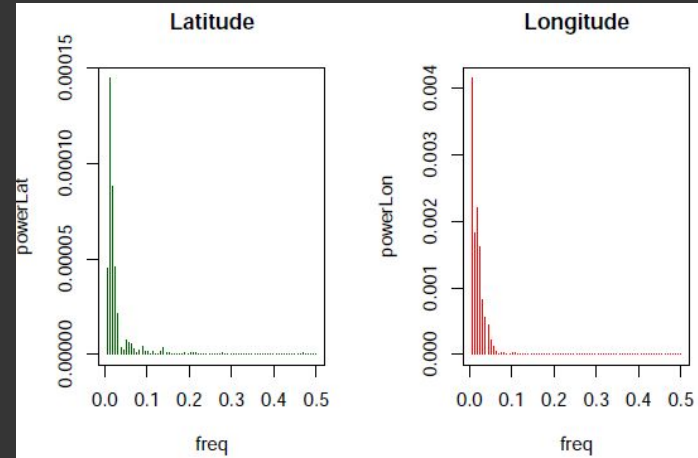
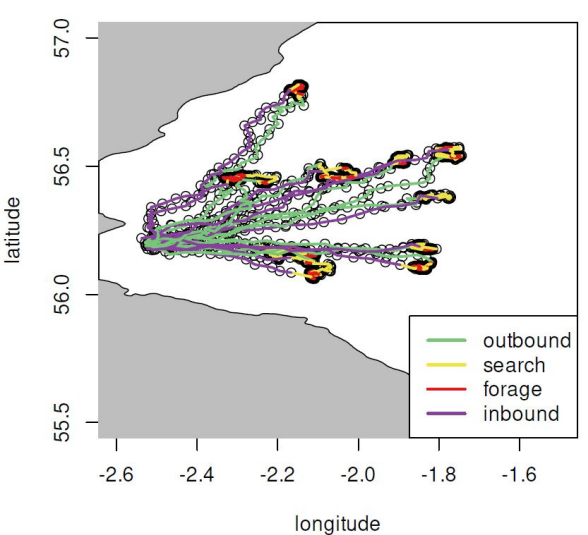
— NSST application



– Nyquist-Shannon sampling theorem (NSST)

NSST states that “If a function $x(t)$ contains no frequencies higher than B hertz, then it can be completely determined from its ordinates at a sequence of points spaced less than $1/(2B = \text{Nyquist rate})$ seconds apart.”

$$\delta t < 1/2B$$

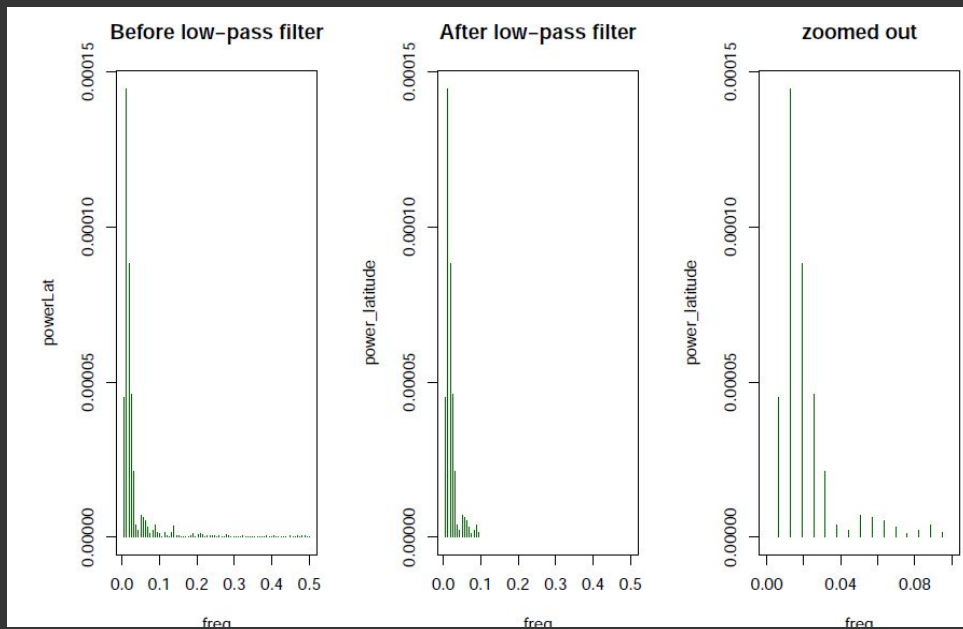


— NSST application

$B = f_s/2$ or $B = 0.5$ in the normalized spectrum, (the signal occupies the whole spectrum), in practice most of the energy of the signal is at low frequencies

One can decide to discard high frequencies at a $B' < f_s/2$.

Then subsample, working at a $f_s' = 2B' < f_s$,



— References

Akeresola et al. (2024). Validating hidden Markov models for seabird behavioural inference, *Ecology and Evolution*. doi/10.1002/ece3.11116

Nathan et al. (2022). Big-data approaches lead to an increased understanding of the ecology of animal movement. *Science*. 375(6582). doi:10.1126/science.abg1780

Thanks for your attention

